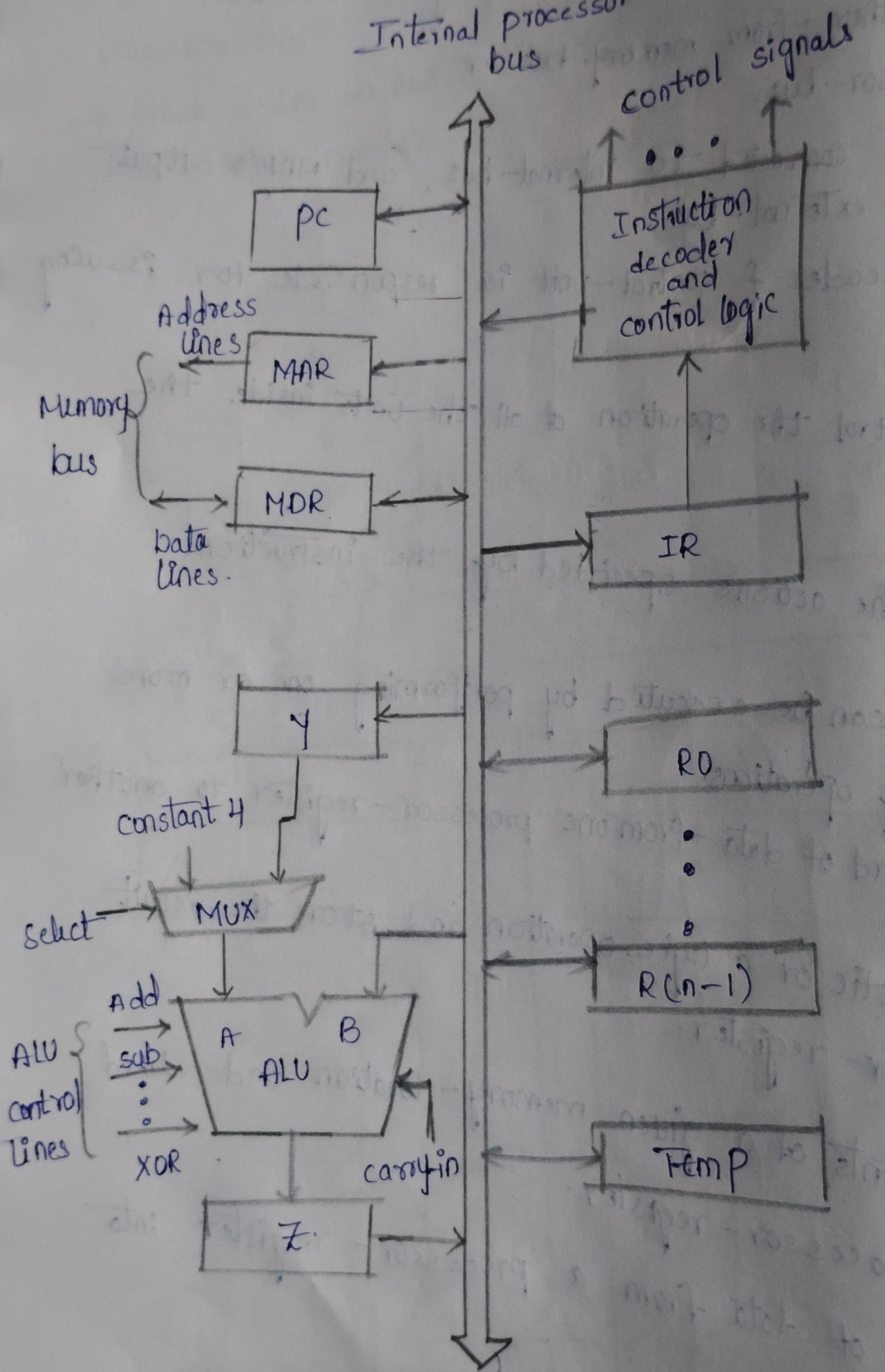
21) Explain the working of single bus organization of datapath?

Ans:-

→ MDR has 2 inputs and 2 outputs. Data may be loaded
→ into MDR either from memory-bus or
→ from processor-bus.
→ MAR's input is connected to internal-bus, and MAR's output is connected to external-bus.
→ Instruction decoder & control-unit is responsible for issuing the signals.
→ That control the operation of all the units inside the processor.
→ implementing the actions specified by the instruction.

An instruction can be executed by performing one or more of the following operations.

→ Transfer a word of data from one processor-register to another or to the ALU.
→ perform arithmetic or a logic operation and store the result in a processor-register.
→ Fetch the contents of a given memory-location and load them into a processor-register.
→ store a word of data from a processor-register into a given memory-location



Single-bus organization of the datapath inside a processor

(20) Analyze how execution of a complete instruction carries out.

→ consider the instruction Add (R_3), R_1 which adds the contents of a memory-location pointed by R_3 to register R_1.

→ Executing this instruction requires the following actions:

1) Fetch the instruction
2) Fetch the first operand
3) perform the addition.
4) Load the result into R_1

→ control sequence for execution of this instruction is as follows

1) Pcount, MARin, Read, Select 4, Add, Zin
2) Zout, PCin, Yin, WMFC
3) MDRout, IRin
4) R3out, MARin, Read
5) R1out, Yin, WMFC
6) MDRout, Select Y, Add, Zin
7) Zout, R1in, End.

Instruction execution proceeds as follows:

Step1 → The instruction-fetch operation is initiated by loading contents of PC into MAR & sending a Read request to Memory.

Step2 → Updated value in Z is moved to PC.

Step3 → fetched instruction is moved into MDR and then to IR

· Step 4 → contents of R_3 are loaded into MAR

Step 5 → contents of R_1 are transferred to y to prepare for addition.

Step 6 → when Read operation is completed

Step 7 → sum is stored in z. then transferred to R_1.

(Q3) Explain the execution of branch instruction?

Ans: control sequence for an unconditional branch instruction is as follows:

1) pcount, MARin, Read, Select 4, Add, Zin

2) Zout, PCin, Yin, WMFC

3) MDRout, IRin

4) offset-field-of-IRout, Add, Zin

5) Zout, PCin, End.

→ The processing starts, as usual, the fetch phase ends in step 3.

→ In step 4, the offset-value of pc is already available in register y, the offset x is gated onto the bus, and an addition operation is performed.

→ In step 5, the result, which is the branch-address, is loaded into the PC.

→ The offset x used in a branch instruction is usually the difference b/w the branch target-address and the address immediately following the branch instruction.

→ In case of conditional branch, we need to check the status of the condition-codes before loading a new value into the PC.

eg:- offset-field-of-IRout, Add, Zin, If N=0 then End
immediately after step 4- If N=1, step 5 is performed
to load a new value into PC.

(24) Define pipelining? Describe instruction pipelining cycle with example.

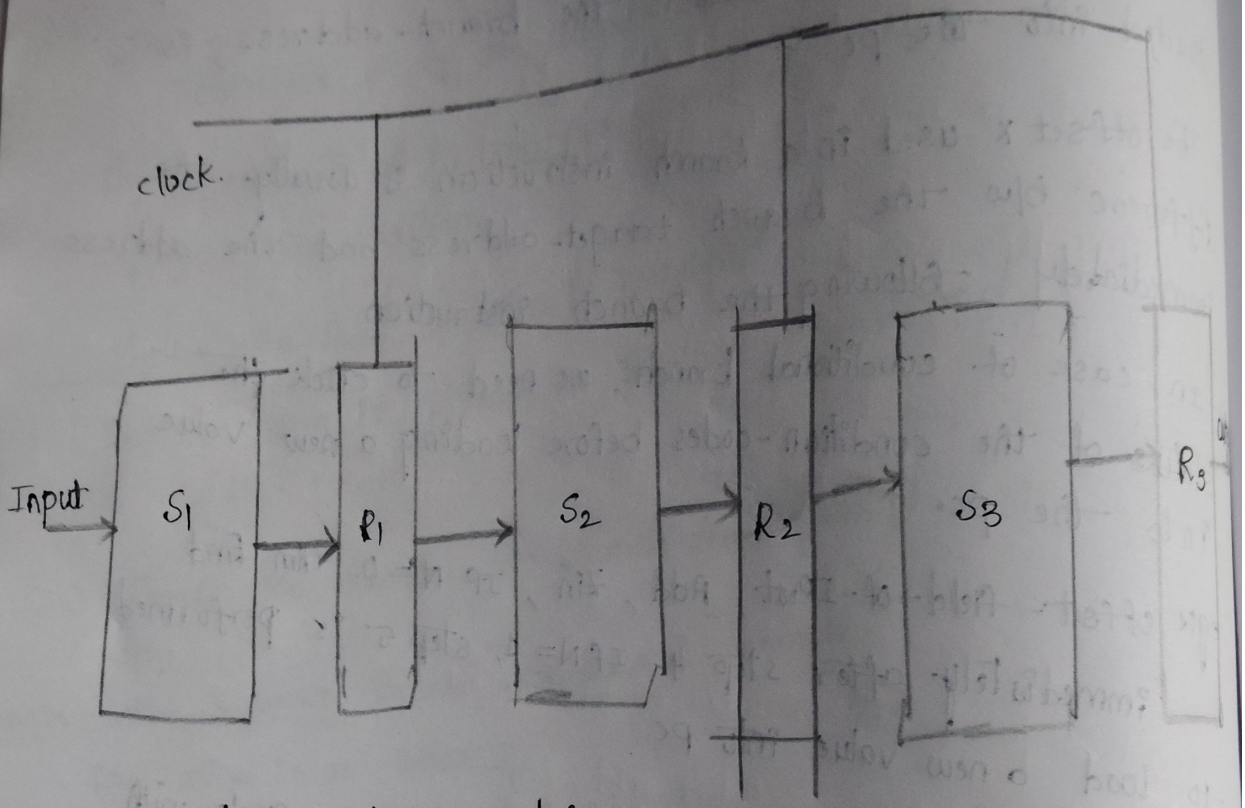
→ pipelining is the process of accumulating instruction from the processor through a pipeline. It allows storing executing instructions in an orderly process. It is also known as pipeline processing.

→ pipeling increases the overall instruction throughput

Types of pipeline
It is divided into 2 categories.

1) Arithmetic pipeline

2) Instruction pipeline



Instruction pipelining cycle is

1. Instruction fetch (IF)
   → The instruction is fetched from memory using the program counter
   → The pc is then incremented to point to the next instruction
2. Instruction Decode (ID)
   → The fetched instruction is decoded
   → The required registers are identified
3. Execute (Ex)
   → The Arithmetic Logic Unit (ALU) performs the required operation
   → for branch instructions, the target address is calculated.

(4) Memory Access (MEM)

→ Data is read from or written to memory if required

→ This stage is mainly used by load and store instructions.

(5) Write Back (WB)

→ The final result is written back to the register file

→ The instruction completes its execution.

Example of Instruction pipelining:

I1: ADD R_1, R_2, R_3

I2: SUB R_4, R_1, R_5

I3: MUL R_6, R_4, R_7

Pipeline Execution Table:-

Clock cycle	Instruction 1	Instruction 2	Instruction 3.
cycle 1	IF	—	—
cycle 2	ID	IF	IF
cycle 3	EX	ID	ID
cycle 4	MEM	EX	EX
cycle 5	WB	MEM	MEM
cycle 6	—	WB	WB.
Cycle 7	—	—	

(25) Write a sequence of control steps to execute the instruction Add $(R_2), R_1$ on single bus architecture.

ADD $(R_2), (R_1)$

→ R_2 out, MARin, Read, WMFC

→ MDRout, Yin

→ R_1 out, MARin, Read, WMFC

→ Zout, R_2 out, MARin, write, WMFC, End.

→ MDRin.